

Functional Requirements Document (FRD)

Module 1: Patient Platform – Appointment Booking & Onboarding

1. Introduction

This module covers all patient-facing interactions for **Accomplish Health**, including appointment scheduling, insurance verification, payment capture, consent forms, and clinical intake questionnaires.

The goal is to provide a seamless, HIPAA-compliant experience for new and existing patients booking appointments, completing intake forms, and confirming visits online.

Key features include:

- Stepwise guided appointment booking flow.
- Insurance and credit card verification.
- Combined consent acknowledgment.
- Digital clinical intake form post-booking.
- Email confirmations and reminders.

The scope includes responsive web application (desktop/mobile), developed using Caspio Low-Code Builder, with integration to Orbit (insurance) and Stripe (payments).

2. Functional Requirements

FRD 1: Basic Details & Demographic Information

ID	Description	Backend Notes
1.1	The system shall display the Appointment Booking form when the user clicks “Schedule Your Appointment.”	Form embedded in webpage via Caspio DataPage.

1.2	The system shall collect user details including: First Name, Last Name, Address, City, State, Postal Code, Email, and Phone Number.	Stored in Caspio patient table.
1.3	The system shall allow users to select Gender and Date of Birth.	Data validation: DOB cannot exceed current date.
1.4	The system shall provide fields for Referring Physician and Partner.	Optional fields; stored if entered.
1.5	The system shall validate all required fields before moving to next step.	Frontend JS validation via Caspio rules.
1.6	The system shall display progress indicator ("Basic Details → Choose Appointment → ...").	Configured through multi-step form flow.

FRD 2: Choose Appointment (Doctor & Slot Selection)

ID	Description	Backend Notes
2.1	The system shall prioritize showing the <i>first available appointment</i> before specific provider selection.	Query sorted by earliest date/time.
2.2	Users can optionally select a preferred clinician or dietitian.	Drop-down populated from provider list.
2.3	Calendar view shall be visible above appointment slots.	Inline calendar component via Caspio JS integration.
2.4	Available slots are displayed dynamically by selected date.	Backend logic filters provider availability table.
2.5	"Next Step" button shall only enable once an appointment slot is selected.	JS validation event-driven.

FRD 3: Insurance Information

ID	Description	Backend Notes
3.1	The system shall display an option: "Use My Insurance" or "Self Pay."	Boolean flag stored per booking.

3.2	For “Use My Insurance,” user shall upload a photo of the insurance card (front/back).	File upload field saved to secure Caspio file storage.
3.3	The system shall validate insurance through Orbit API (eligibility check).	API: POST /orbit/verify-insurance.
3.4	For “Self Pay,” the system shall display a message with self-pay terms and checkbox to confirm.	Checkbox required for continuation.
3.5	The system shall allow user to proceed to payment once insurance validation or self-pay confirmation is complete.	Validation flag = true required.

FRD 4: Payment & Credit Card Verification

ID	Description	Backend Notes
4.1	The system shall require entry of credit card details for appointment confirmation.	Stripe Checkout integration.
4.2	Required fields: Cardholder Name, Card Number, Expiry Date, CVV, Billing Address.	Tokenization handled by Stripe; no raw card data stored.
4.3	The system shall include a message explaining “Why credit card is required” with a link to Cancellation Policy.	Static text field configurable in CMS.
4.4	The system shall verify payment token validity before allowing continuation.	API: POST /stripe/verify-token.
4.5	The system shall store payment confirmation ID linked to booking record.	DB: appointment.payment_token_id.

FRD 5: Consent Forms

ID	Description	Backend Notes
5.1	The system shall consolidate all consent forms into one acknowledgment checkbox.	Static content box with combined text.
5.2	Consent text shall include: Information Accuracy, Informed Consent for Patient Monitoring, Care Management, and Privacy Agreement.	Fetches from Consent Text Table.

5.3	The system shall require the user to check “I agree” before proceeding.	Validation: Checkbox required.
5.4	The system shall store timestamp and user’s name as digital signature confirmation.	DB: consent_accepted_at + patient_id.

FRD 6: Appointment Confirmation

ID	Description	Backend Notes
6.1	The system shall display a “Confirm Appointment” modal summarizing all details.	Appointment summary from session state.
6.2	Upon confirmation, the system shall generate appointment record and send confirmation email.	API: POST /appointment/create.
6.3	The confirmation email shall contain appointment date/time, provider name, and a link to patient intake form.	Email template stored in Caspio Mail Settings.
6.4	The system shall show a success message: “Appointment booked successfully.”	On-screen modal message.

FRD 7: Clinical Intake Form

ID	Description	Backend Notes
7.1	The system shall provide access to the intake form immediately post-booking or via email link.	URL tokenized per patient.
7.2	The form shall include sections: Medical History, Family History, Surgical History, GYN, Medication, Allergies, Weight, Functional Status, Social, Nutrition.	Stored across related tables.
7.3	Each section shall be navigable via step progress bar.	JS-driven stepper component.
7.4	The system shall allow saving progress and resuming later.	Session tokens for draft saving.
7.5	The system shall send reminders via email if intake form remains incomplete after 24 hours.	Scheduled job in Caspio triggers email event.

7.6	Optional skip allowed initially, but completion required before appointment date.	Flag: intake_status = PENDING/COMPLETE.
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FRD 8: Notifications & Messaging

ID	Description	Backend Notes
8.1	The system shall send email confirmation immediately after booking.	Email service: Caspio Mailer / SMTP.
8.2	The system shall send reminder emails 24 hours before appointment.	Automated scheduler.
8.3	The system shall send an alert for insurance verification failure.	Event trigger linked to insurance_status = "Failed."
8.4	The system shall show inline notification on page load if previous booking exists.	DB query check for active appointment.

3. User Journeys

Journey A: New Patient Appointment Booking

1. User clicks "Schedule Your Appointment."
2. Enters basic details and demographics.
3. Selects *first available appointment*.
4. Uploads insurance card or selects *Self Pay*.
5. Enters credit card details and agrees to terms.
6. Reviews confirmation modal and confirms.
7. Receives confirmation email with intake form link.

Journey B: Returning Patient

1. System detects existing patient by email or phone.
2. Pre-fills demographic and insurance data.
3. User updates or confirms details.
4. Proceeds with appointment and payment confirmation.
5. Receives confirmation and optional intake form (if new info required).

Journey C: Intake Form Completion

1. User clicks intake form link from email.
2. Completes all required sections.
3. Reviews and submits.
4. System marks intake as “Completed.”
5. Provider notified through admin dashboard.

4. User Stories & Acceptance Criteria (GWT Format)

User Story 1 – Basic Information Entry

As a patient,

I want to enter my personal and demographic details,

so that I can begin my appointment booking process and ensure my information is correctly recorded.

Acceptance Criteria:

Given the patient is on the Basic Details screen,

When they enter all required fields such as First Name, Last Name, Email, Phone, Address, Gender, and Date of Birth,

Then the system shall validate the data for correctness and completeness before allowing the user to proceed.

Given an invalid input (e.g., missing email or incorrect date format),

When the user tries to continue,

Then the system shall display an inline error message and prevent progression until corrections are made.

User Story 2 – Appointment Selection

As a patient,

I want to view and select from the first available appointment slots or a preferred provider,
so that I can schedule my consultation at a convenient time.

Acceptance Criteria:

Given that the patient has entered valid demographic details,

When the system loads the appointment selection screen,

Then the system shall automatically display the first available appointments based on provider availability and time zone.

Given that the patient chooses a preferred provider,

When the provider's schedule is loaded,

Then the system shall show available time slots for that specific provider and highlight the selected slot when chosen.

Given that the selected slot becomes unavailable,

When the patient attempts to proceed,

Then the system shall notify the user and prompt them to select another available time.

User Story 3 – Insurance Upload or Self-Pay

As a patient,

I want to either upload my insurance details or choose a self-pay option,
so that billing can be processed appropriately for my appointment.

Acceptance Criteria:

Given that the patient selects "Use My Insurance,"

When they upload photos of the insurance card (front and back),

Then the system shall validate the uploaded files and send the data to the Orbit API for eligibility verification.

Given that insurance verification succeeds,
When the response is received from the API,
Then the system shall mark the insurance as “Verified” and allow the user to proceed to payment.

Given that insurance verification fails,
When the API returns an error or invalid response,
Then the system shall display “Insurance could not be verified” and offer the option to proceed as “Self Pay.”

Given that the patient selects “Self Pay,”
When the confirmation checkbox for self-pay terms is ticked,
Then the system shall mark the booking as “Self Pay” and allow the user to continue.

User Story 4 – Payment Verification

As a patient,
I want to securely provide my credit card details,
so that my appointment can be confirmed and any cancellation policy requirements can be enforced.

Acceptance Criteria:

Given that the patient has completed the insurance/self-pay step,
When they reach the payment page,
Then the system shall display credit card input fields for Cardholder Name, Card Number, Expiry Date, CVV, and Billing Address.

Given that valid card details are entered,
When the user clicks “Next Step,”
Then the system shall send the data to Stripe for tokenization and validation.

Given that the Stripe API confirms the token,
When the system receives a valid token,
Then the booking is updated with the payment reference and the patient can continue to the consent step.

Given that the token verification fails,
When the response indicates an error,
Then the system shall display an inline message “Payment verification failed. Please try again.”

User Story 5 – Consent Acknowledgment

As a patient,
I want to review and agree to all required consent forms in one step,
so that my personal data and medical information can be legally processed for care.

Acceptance Criteria:

Given that the patient is on the Consent screen,
When the system displays the consolidated consent content (covering Accuracy of Information, Care Management, Monitoring, and Privacy Agreement),
Then the user must scroll through the content to enable the “I Agree” checkbox.

Given that the patient selects “I Agree,”
When they proceed to the next step,
Then the system shall store the consent response with timestamp, user name, and IP address as digital acknowledgment.

Given that the patient does not agree,
When they try to continue without checking the box,
Then the system shall block the next step and show “You must agree to continue.”

User Story 6 – Appointment Confirmation

As a patient,
I want to review all appointment details before finalizing,
so that I can verify accuracy and confirm my booking with confidence.

Acceptance Criteria:

Given that all prior steps are complete,
When the system loads the confirmation modal,
Then it shall display a summary including Name, Appointment Date, Provider, Insurance Type, and Payment Status.

Given that the patient confirms,
When they click “Confirm Appointment,”
Then the system shall create the booking record, send a confirmation email, and display a “Booking Successful” message.

Given that the confirmation email is sent,
When the patient checks their inbox,
Then they shall receive an email with booking summary and a link to the clinical intake form.

User Story 7 – Clinical Intake Completion

As a patient,

I want to complete my medical and lifestyle intake forms online,
so that my provider can review my health information prior to my appointment.

Acceptance Criteria:

Given that the patient clicks the intake link from the confirmation email or screen,

When the intake form opens,

Then the system shall display a step-by-step form with sections for Medical, Family, Surgical, GYN, Medications, Allergies, Weight, Functional, Social, and Nutrition history.

Given that the patient fills in each section,

When the “Next” button is clicked,

Then the system shall save progress automatically for that section.

Given that the form is incomplete,

When 24 hours have passed since booking,

Then the system shall send a reminder email prompting completion.

Given that the patient completes and submits the form,

When all required sections are filled,

Then the intake status is updated to “Complete” and the provider dashboard is automatically notified.

User Story 8 – Notifications and Reminders

As a patient,

I want to receive email confirmations and reminders about my appointments and forms,
so that I don’t miss any important actions before my visit.

Acceptance Criteria:

Given that the appointment has been confirmed,

When the booking record is created,

Then the system shall automatically send a confirmation email containing appointment and provider details.

Given that the appointment date is approaching,

When it is 24 hours before the appointment,

Then the system shall send a reminder email with the date and time.

Given that the intake form remains incomplete,
When 24 hours have passed since confirmation,
Then the system shall send a follow-up email prompting completion of the intake form.

Given that insurance verification fails,
When the Orbit API returns an invalid response,
Then the system shall email the patient with instructions to re-upload their insurance or select self-pay.